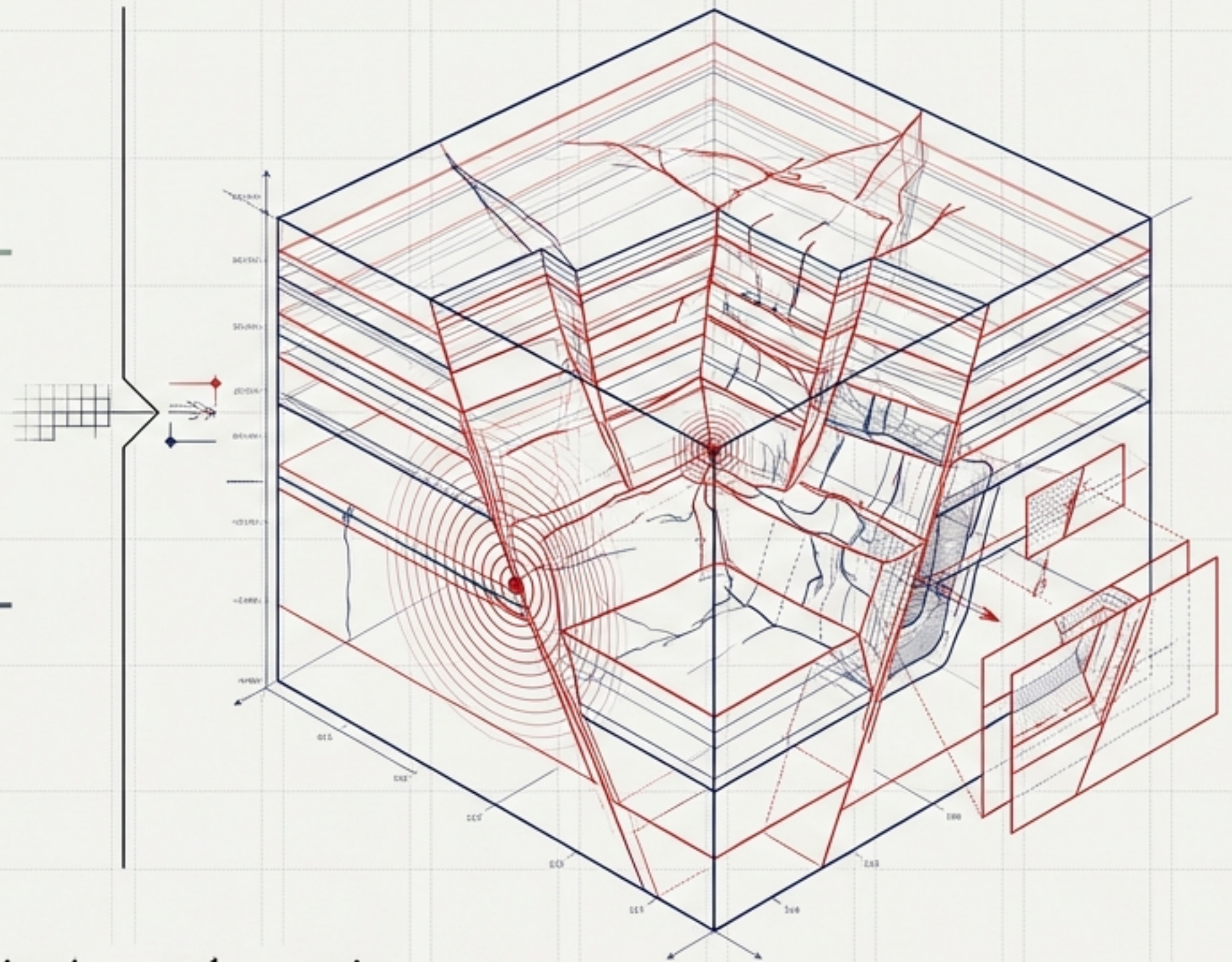
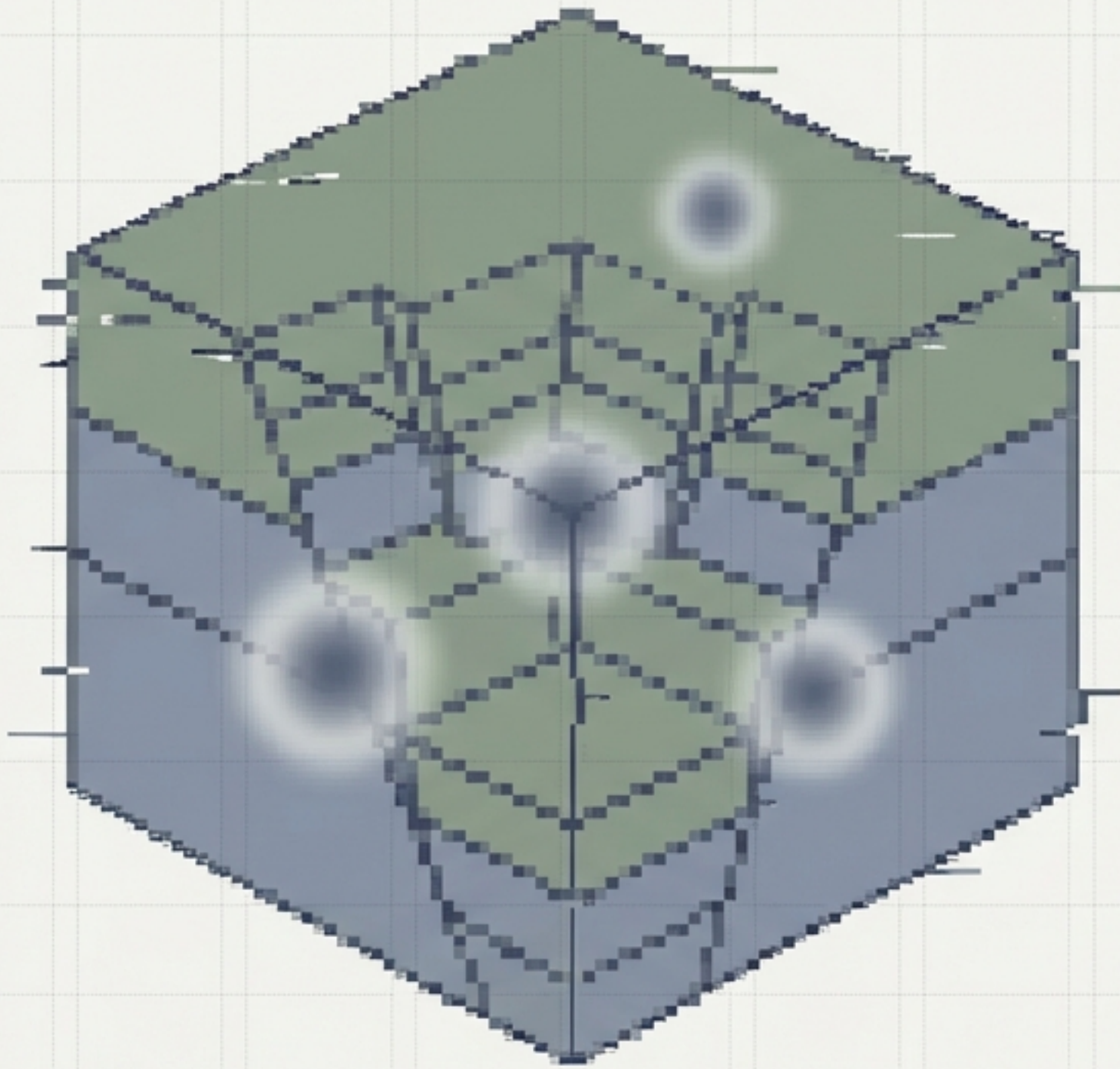


Fractal Tomography & The End of the Seismic Mirage

Resolving the Multi-Decade Paradoxes of Earthquake Geometry
via Precision-Calibrated Bayesian Inference.

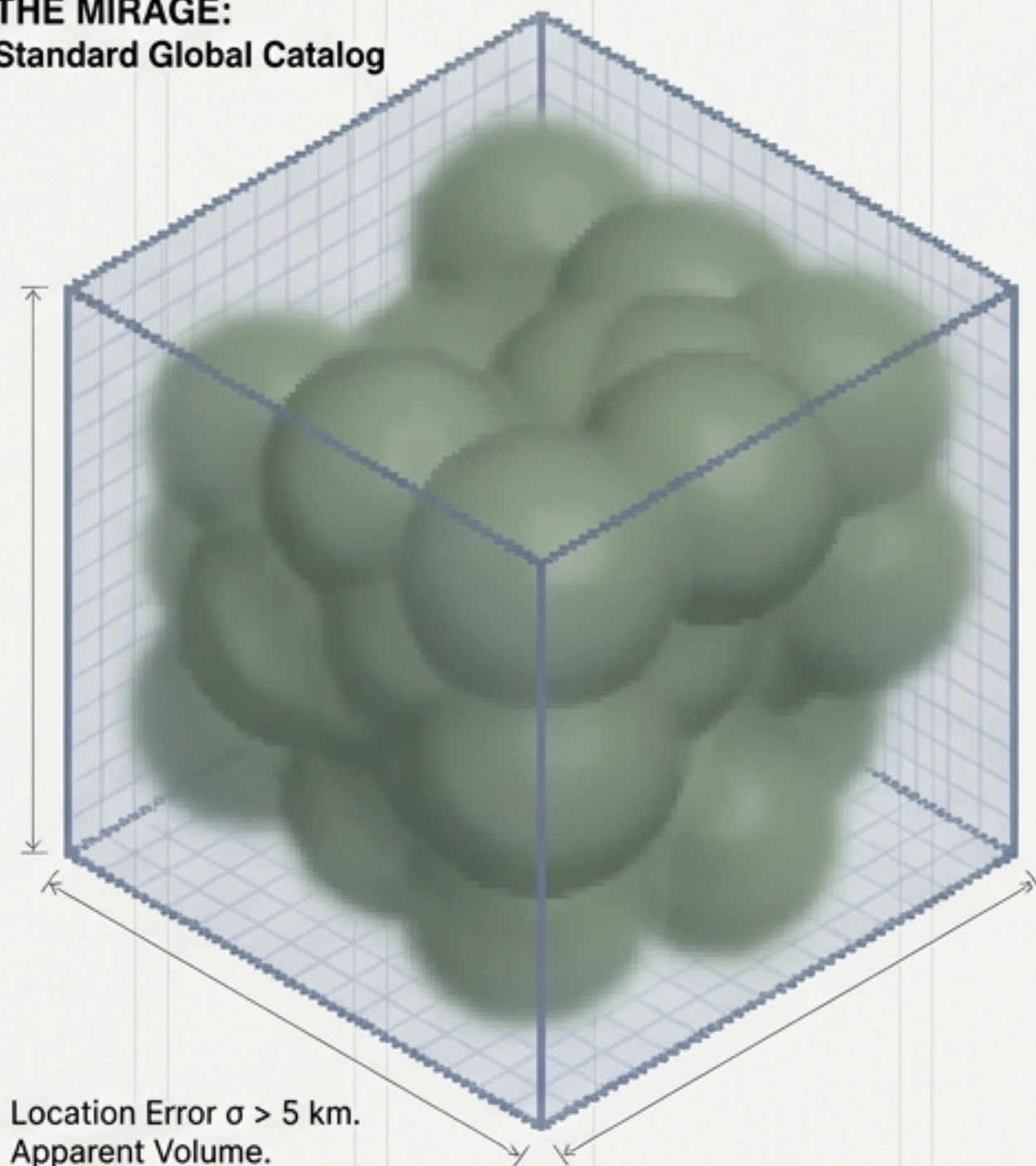


*“Precision is not a luxury; it
is a lens that changes reality.”*

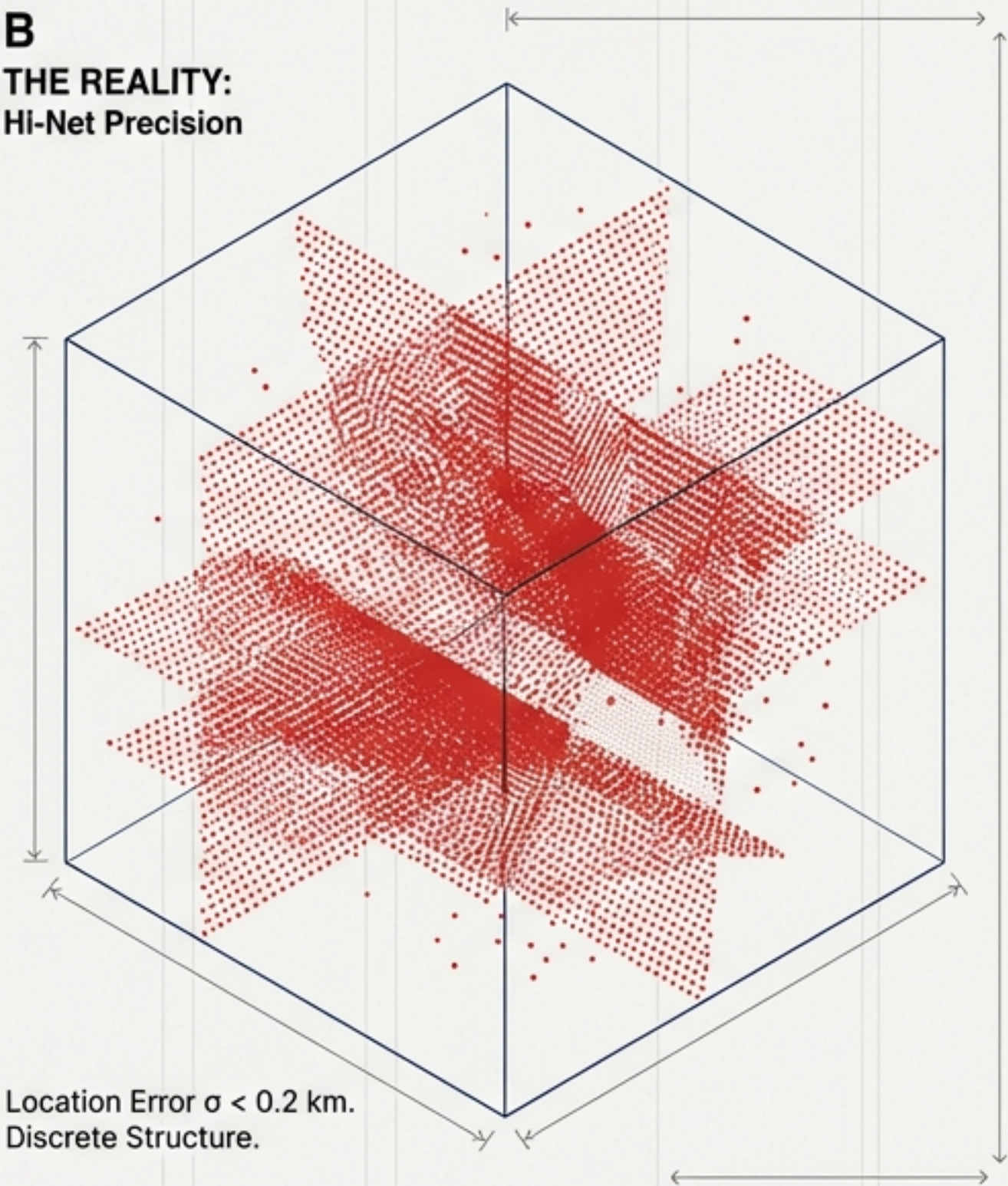
Authors: Firmenich et al. – CEDESUR (Centro de Estudios del Sur)
Based on: “El Marco de Tomografía Fractal y la Arquitectura Sísmica”

For forty years, seismology has been trapped in an 8-bit video game.

A
THE MIRAGE:
Standard Global Catalog



B
THE REALITY:
Hi-Net Precision



When distance resolution is low, a forest looks like a solid green block.

When resolution increases to 4K, the block dissolves into individual trees and leaves.

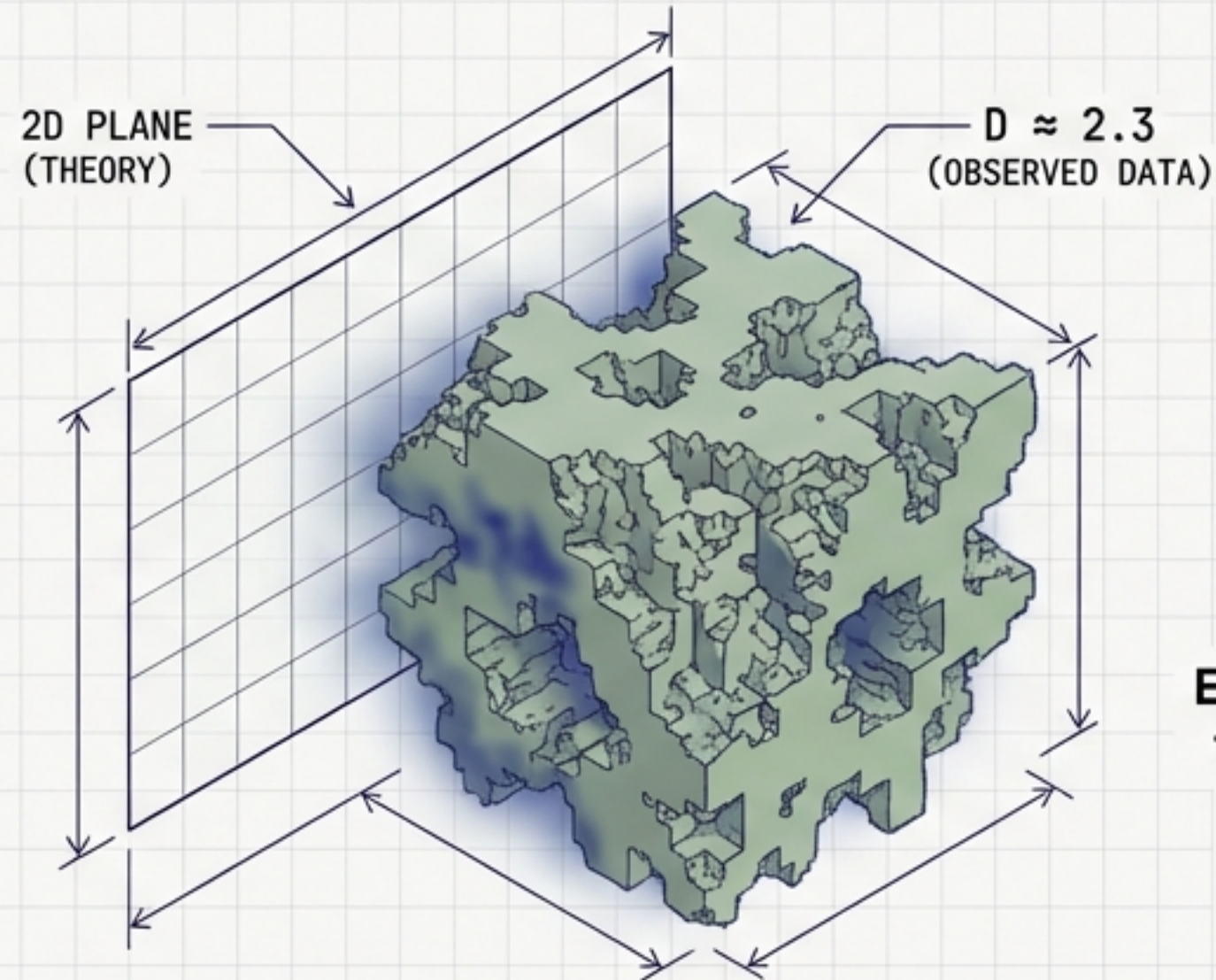
The “solid volume” of earthquakes was an artifact of low resolution.

The new research represents the “4K Remaster” of our understanding of the Earth's interior.

Two historical paradoxes suggested our physics and our math were at war.

The Geometric Paradox

Geologists establish faults as 2D planes (cracks). Yet, global data consistently showed a fractal dimension of $D \approx 2.3$. Why did the Earth appear as a “dimensional monster”—neither a perfect plane nor a solid block?

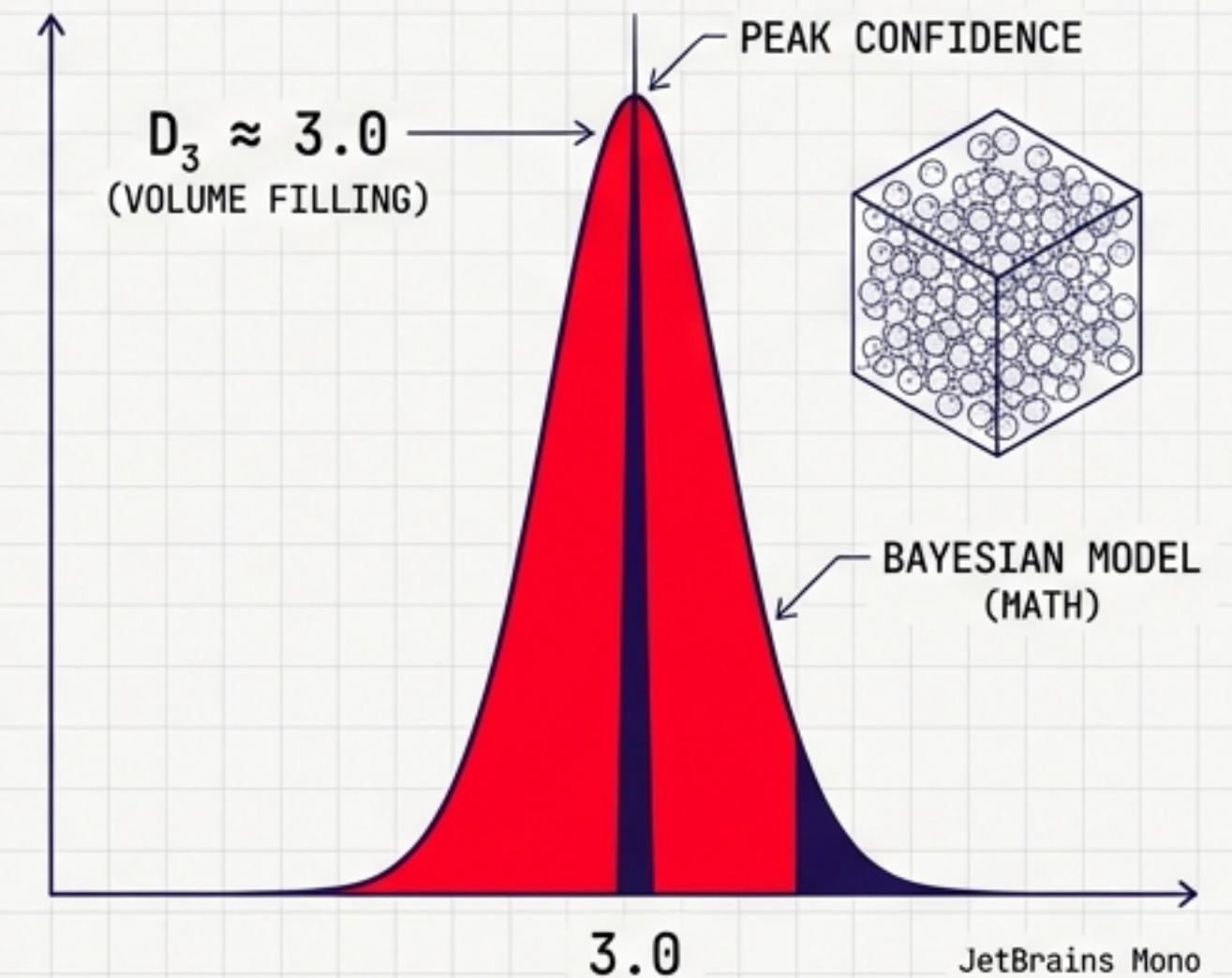


vs.

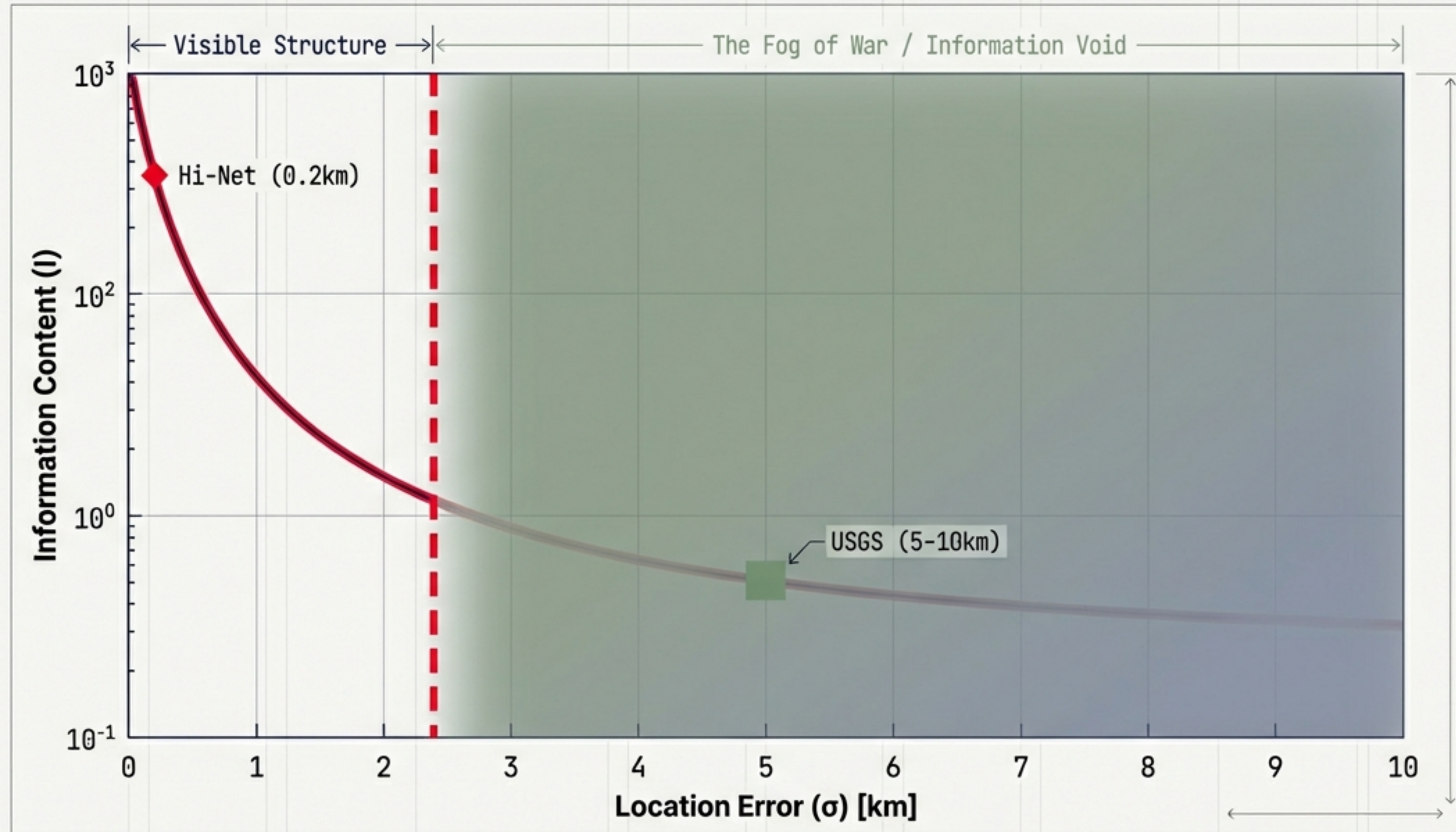
The Conflict:
Either tectonic theory was fundamentally wrong, or the most advanced mathematical models were hallucinating.

The Bayesian Paradox

Advanced Bayesian algorithms analyzing global catalogs (USGS) confidently concluded $D_3 \approx 3.0$. They claimed earthquakes fill volume like bubbles in a sponge.



The Fisher Information Barrier (σ_c) defines the threshold of scientific blindness.



Just as a “fog of war” in strategy games hides the map, there is a mathematical limit to what we can see.

The study derives this limit from first principles:

$$\sigma_c = 2.3 \pm 0.4 \text{ km.}$$

If an instrument's location error exceeds this, the observer enters the zone of ignorance.

Bayesian Saturation: When the algorithm hallucinates volume.

INPUT

Blurry Data ($\sigma > 5\text{km}$)



JetBrains Mono
Data is blurred and
indistinct.

THE GLITCH

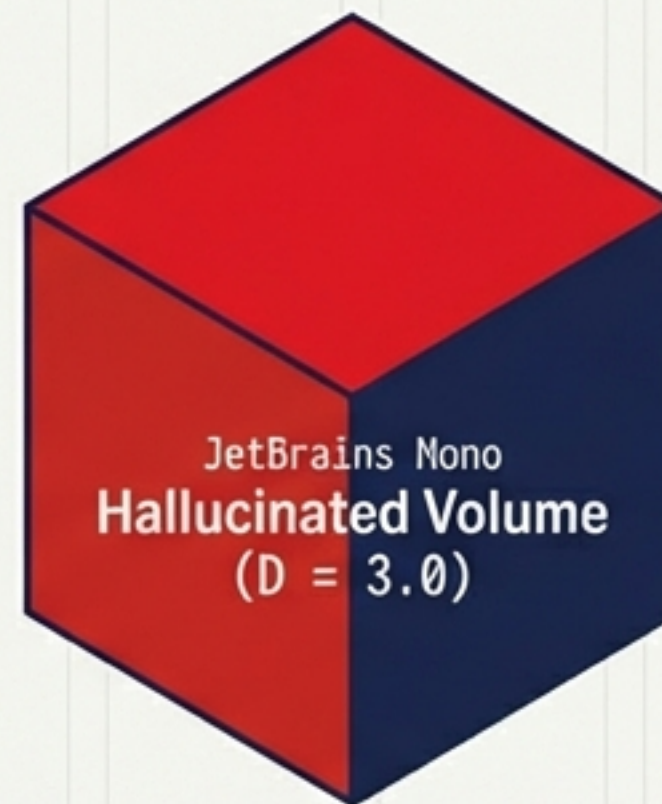
BAYESIAN ALGORITHM

Helvetica Now Display



JetBrains Mono
Hits Fisher Barrier →
Ignores Data →
Reverts to "Priors"

OUTPUT



JetBrains Mono
Hallucinated Volume
($D = 3.0$)

FALSE POSITIVE

Helvetica Now Display

When data is blurry ($\sigma > 2.3 \text{ km}$), the Bayesian algorithm cannot find structure. In the absence of clear data, it "panics" and reverts to its priors (assumptions). Since it assumes space is volumetric, it "hallucinates" a 3D structure to satisfy itself.

"It is like using an AI filter on a blurry photo; the AI invents details that do not exist."

Changing the lens: The hyper-precision of the Hi-Net array

To solve the paradox, the authors abandoned global averages and utilized Japan's Hi-Net system—a “telescope pointing down.” This moves us from the Fog of War to crystalline clarity.

RESOLUTION COMPARISON

Standard Network (USGS)

Hi-Net (Japan)

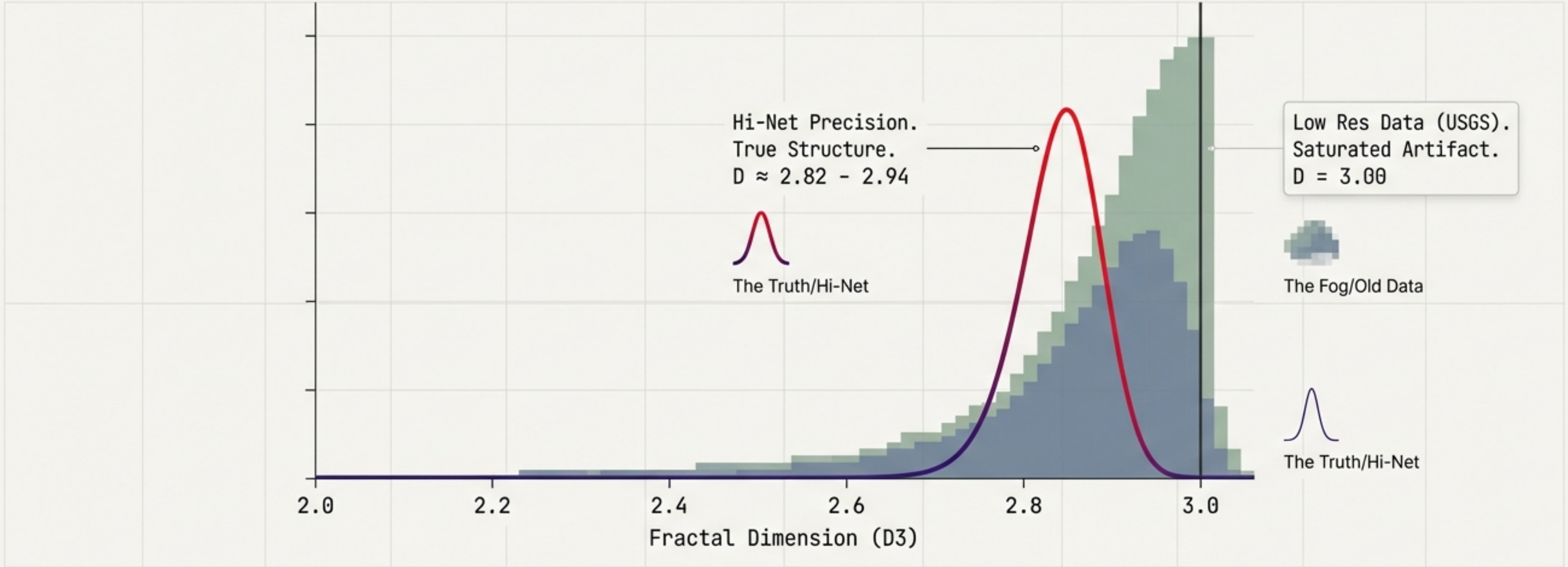
Error ≈ 8 km

Error < 0.2 km

70x Improvement in Resolution



High-definition data shatters the volumetric illusion.

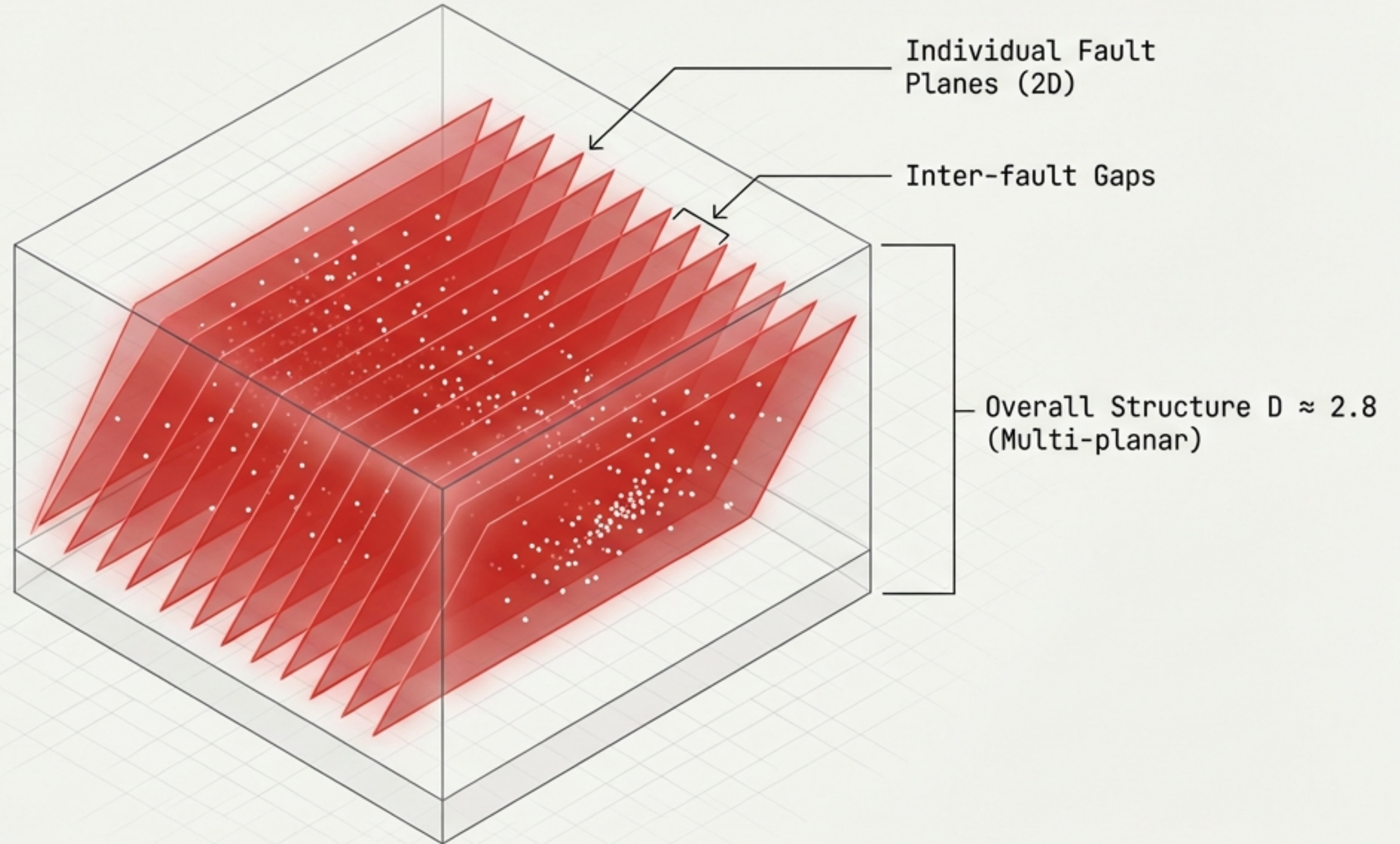


Comparing the 'blurry' Pan-American data against the 'sharp' Hi-Net data reveals the truth. The Low Res data crashes into the 'Bayesian Saturation' barrier ($D=3.0$). The Hi-Net data breaks free, revealing that the true dimension is fractal, complex, and sub-volumetric.

The Reality: The 'Deck of Cards' Architecture

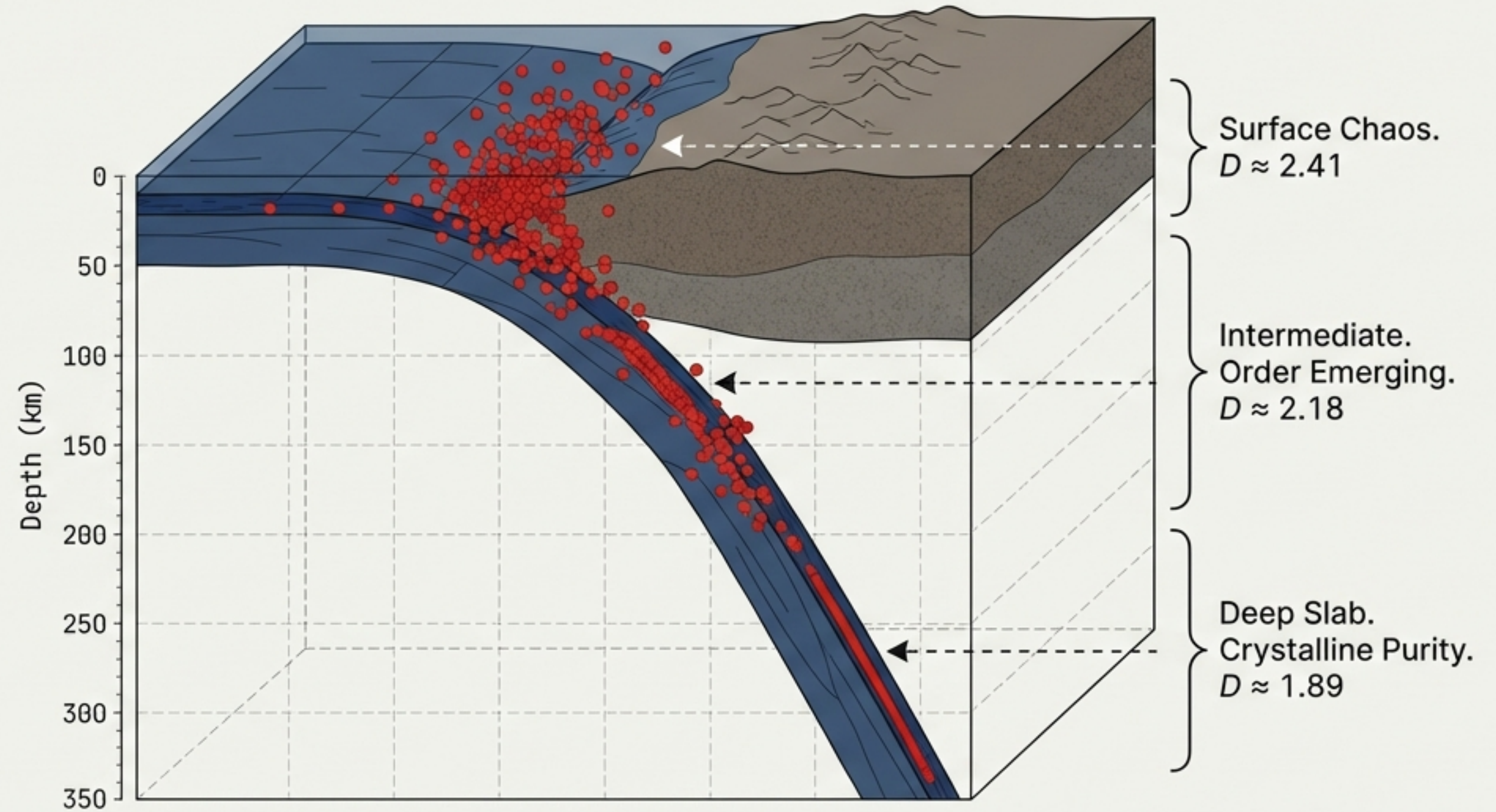
Earthquakes do not fill the rock volume. They organize on a complex stack of multi-planar structures.

Imagine a deck of cards frozen in mid-air. From a distance (Low Res), the stack looks like a solid block. Up close (Hi-Net), you see the gaps between the cards. It is not chaos; it is a complex, sliced order.

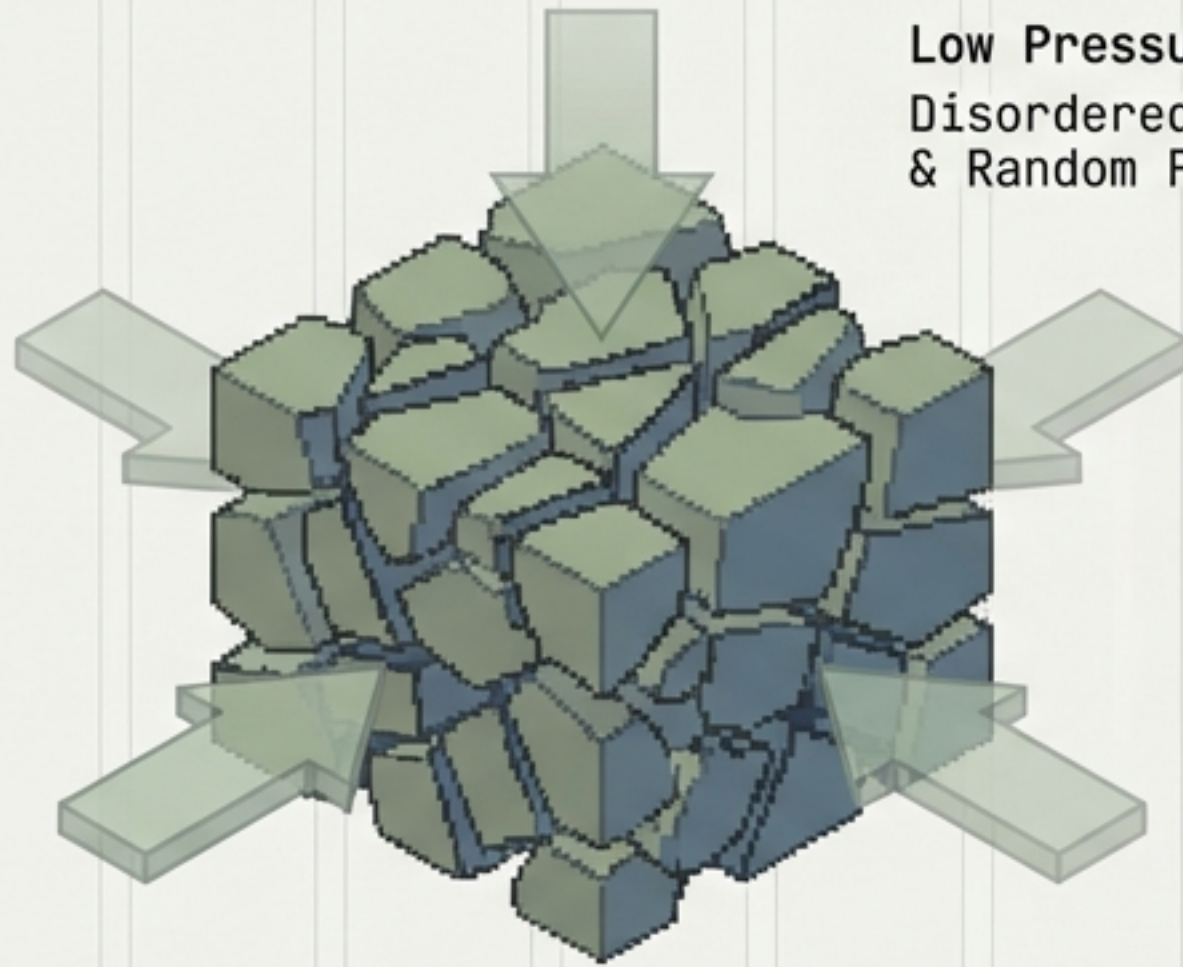


A Journey to the Deep: Chaos planarizes into crystalline order.

Dimension is not constant. As we go deeper into the Earth, the pressure "irons out" the chaos. The structure transitions from a shattered surface to a pure, planar geometry in the deep slab.



The Hydraulic Press: Why the Earth flattens with depth



Low Pressure
Disordered Matrix
& Random Fractures

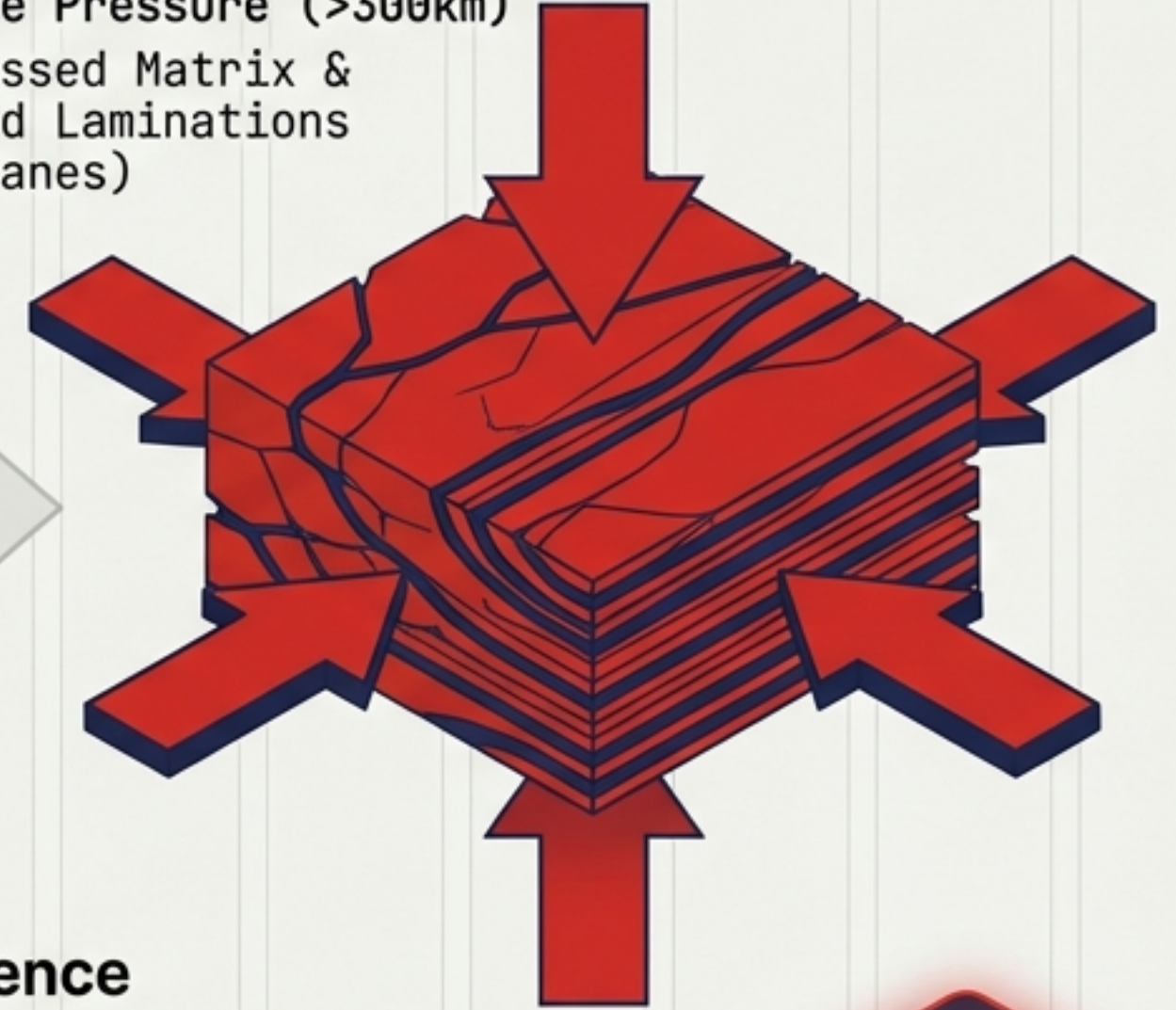
Old Theory

~~Volumetric Expansion.~~
Heat makes structure "gooey"
and blob-like.



Dehydration of
minerals acts like
a cosmic iron.

Extreme Pressure (>300km)
Compressed Matrix &
Aligned Laminations
(2D Planes)

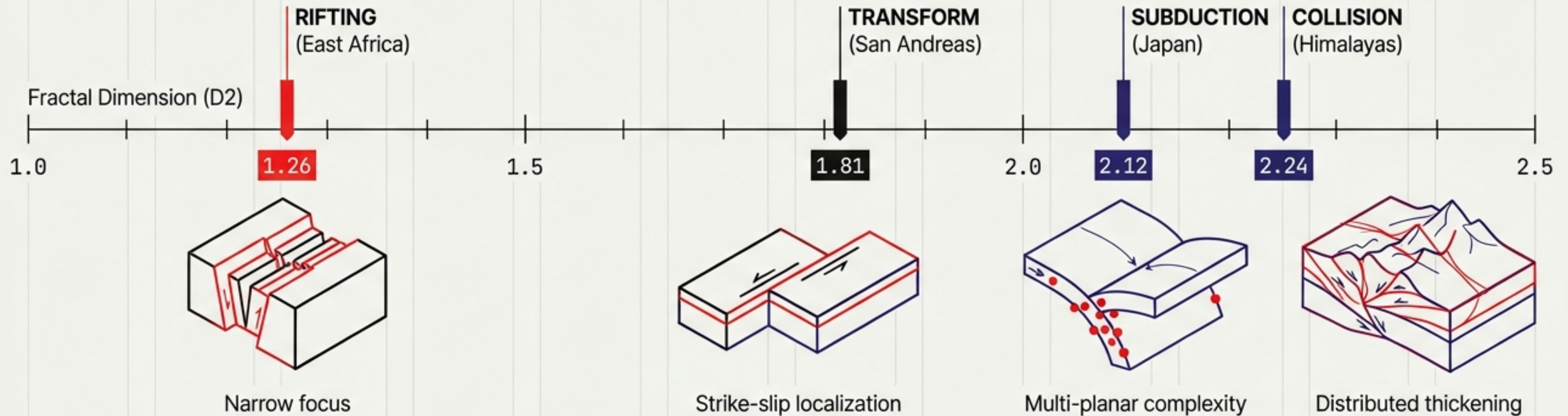


New Evidence

Progressive Planarization.
Pressure forces earthquakes
onto strictly defined 2D planes
(metastable olivine wedges).



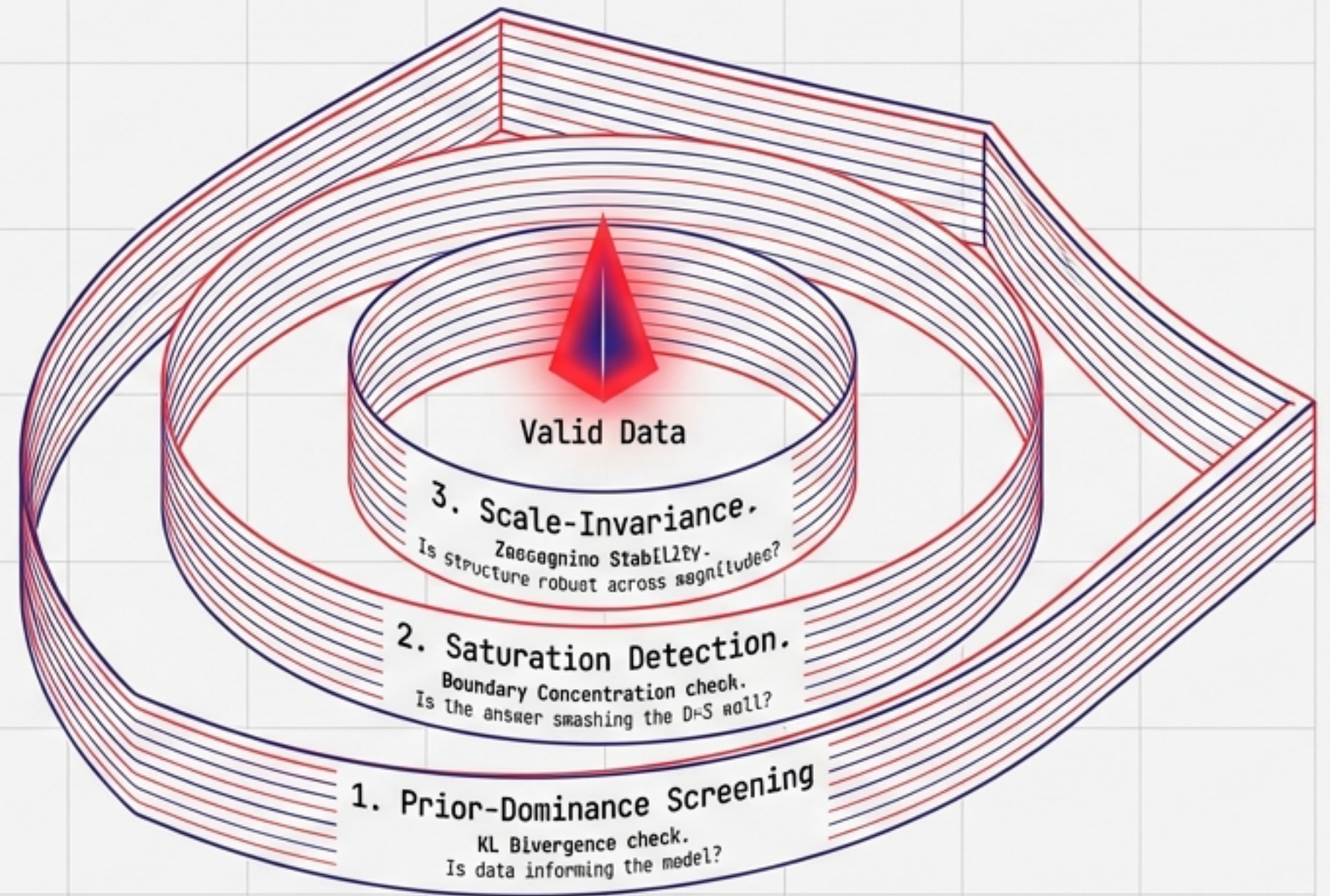
The Tectonic Hierarchy: Classifying power through dimension



The study validates a hierarchy of tectonic settings based on their fractal dimension (ANOVA $F=47.3$). We can now classify the 'power' of a tectonic zone by its geometric complexity.

A new standard for reliability: The Triple-Validation Framework.

Single metrics can lie. This framework prevents future errors by requiring a three-layer defense against artifacts. Only data that passes all three gates is considered real.



Summary of the Paradigm Shift

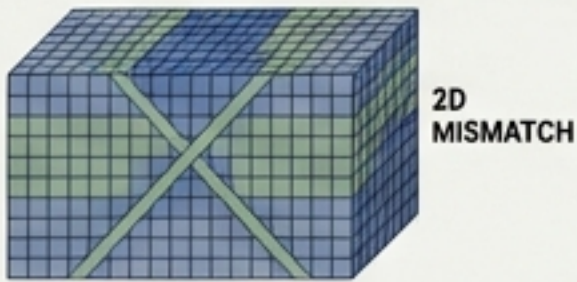
A precise contrast between obsolete assumptions and the newly calibrated, high-resolution reality.

OLD PARADIGM (The Mirage)



The Paradox

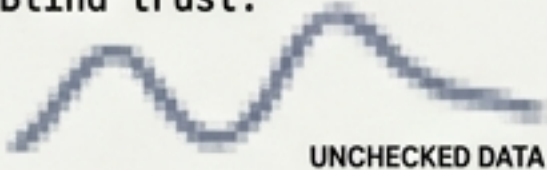
Conflict: 2D Faults vs. 3D Data



Relied on simplified 2D representations, leading to unresolvable conflicts in 3D datasets.

The Limit

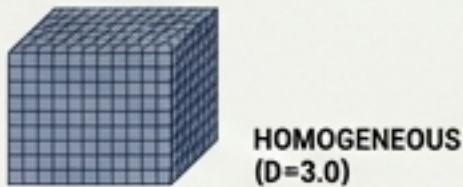
Ignored resolution limits. Blind trust.



Assumed all data points were equally valid, neglecting critical resolution boundaries.

The Structure

Volumetric Sponge (D=3.0). Constant.



modeled structure as a uniform, volume-filling continuum with constant dimensionality.

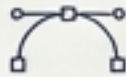
The Method

Catalog-agnostic averages.

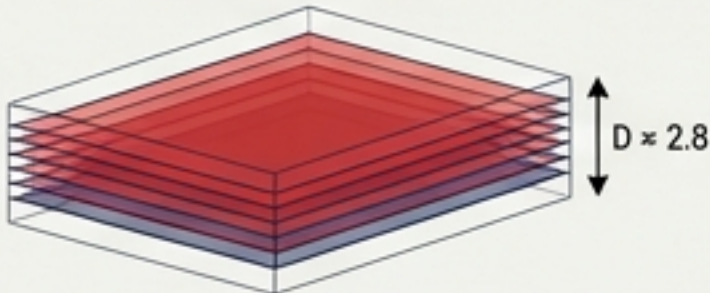


Relied on broad, uncalibrated averages across disparate datasets, masking crucial variations.

NEW PARADIGM (The Reality)

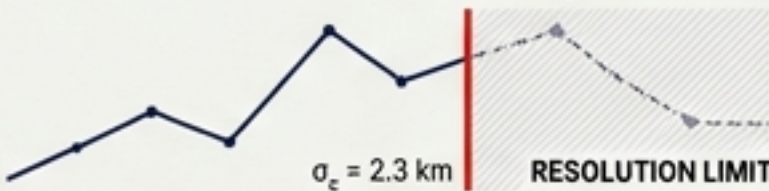


Resolved: "Deck of Cards" Model ($D \approx 2.8$)



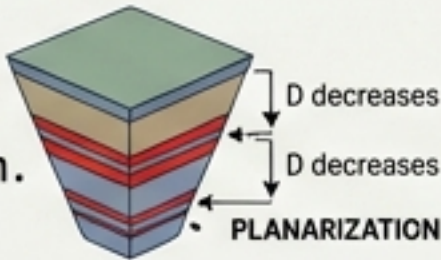
Embraces a fractional dimension model of stacked, planar structures, matching observed 3D complexity.

Fisher Barrier defined at $\sigma_c = 2.3$ km



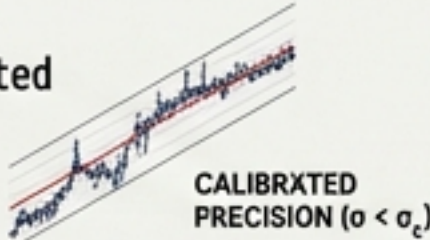
Establishes a strict, empirically determined resolution barrier, discarding unreliable data.

Depth-Dependent Planarization. D drops with depth.



Identifies depth as a critical factor, with structure becoming more planar and dimension decreasing with depth.

Precision-calibrated ($\sigma < \sigma_c$)

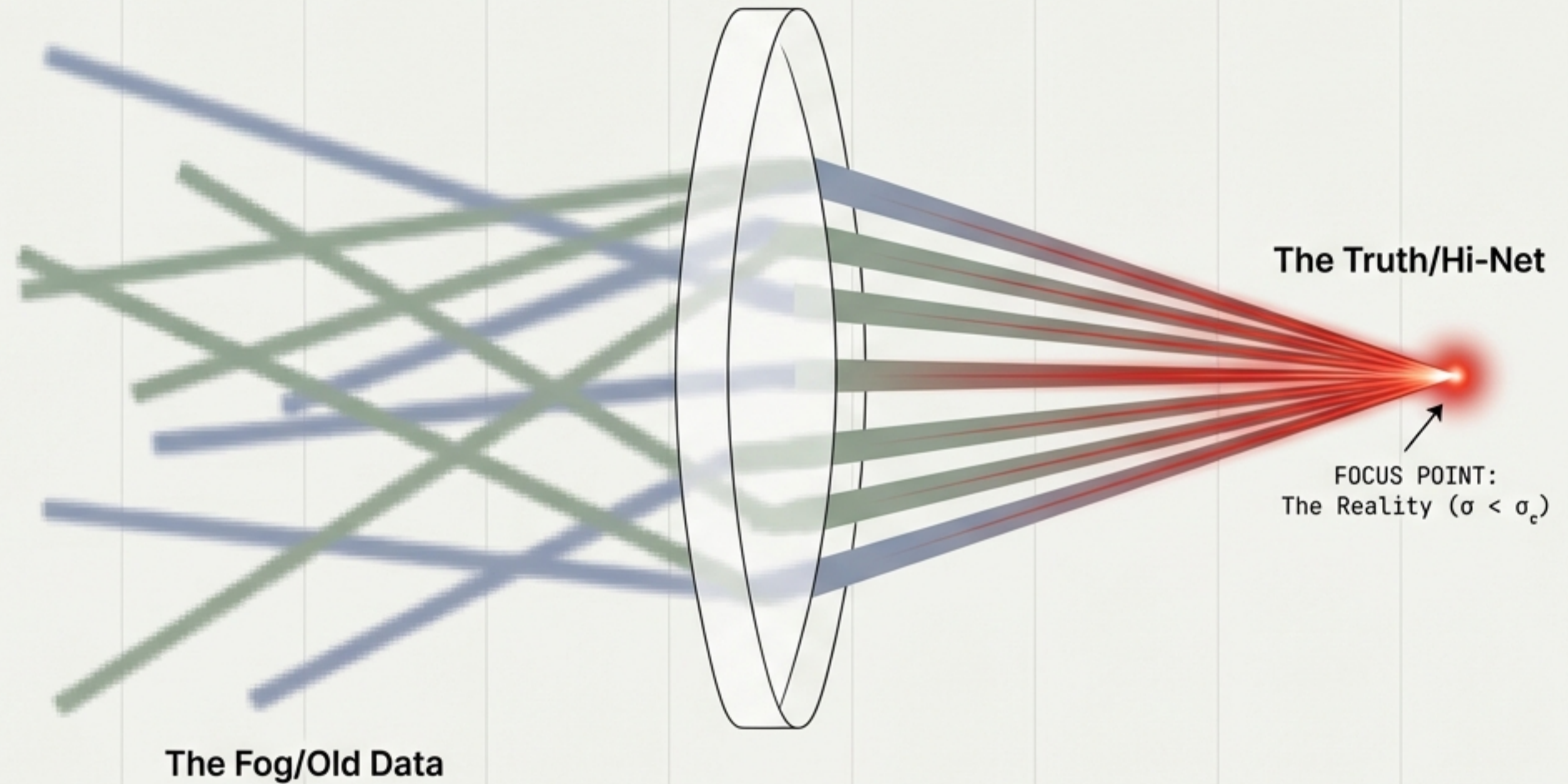


Employs rigorous, catalog-specific calibration against the defined resolution barrier for high fidelity.

Precision is not a luxury; it is a lens that changes reality.

Like the Heisenberg Uncertainty Principle, seismology has a fundamental limit. If we look with instruments that are too blurry ($\sigma > 2.3$ km), we see mathematical ghosts.

The world is not how we *calculate* it with blurry averages. It is how we *see* it when we have the courage—and the technology—to look at the finest details.



Based on: "Fractal Tomography and the Architecture of Seismicity" by Firmenich et al.
Seismic Red / Precision Indigo / Domaine Text / JetBrains Mono